Practical No : 10

Name: Aditya Jadhav

Class: MSC(Com.Sci) Part-1

Roll No : 11

Date:

Q.10) Write a C program for Prim's Algorithm.

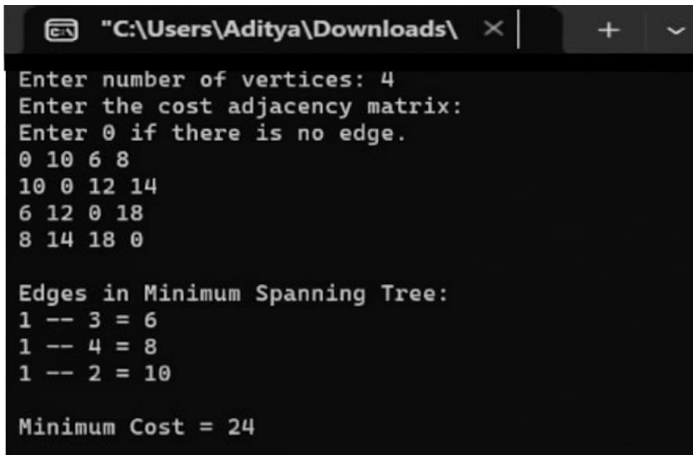
```
#include <stdio.h>
#define MAX 20
#define INF 9999
int main()
{
   int n;
   int graph[MAX][MAX];
   int selected[MAX];
   int i, j;
   int edges = 0;
   int min, x, y;
   int totalCost = 0;
   printf("Enter number of vertices: ");
   scanf("%d", &n);
   printf("Enter the cost adjacency matrix:\n");
   printf("Enter 0 if there is no edge.\n");
   for (i = 0; i < n; i++) {
      for (j = 0; j < n; j++) {
         scanf("%d", &graph[i][j]);
         if (graph[i][j] == 0)
            graph[i][j] = INF;
      }
   }
   for (i = 0; i < n; i++)
      selected[i] = 0;
```

```
    // Start from vertex 0
    selected[0] = 1;
    printf("\nEdges in Minimum Spanning Tree:\n");
    while (edges < n - 1) {
        min = INF;
        x = -1;
        y = -1;
        for (i = 0; i < n; i++) {
            if (selected[i]) {
                for (j = 0; j < n; j++) {
                    if (!selected[j] && graph[i][j] < min) {
                        min = graph[i][j];
                        x = i;
                        y = j;
                    }
                }
            }
        }
        if (x == -1 || y == -1) {
            printf("Graph is not connected.\n");
            return 0;
        }
        printf("%d -- %d = %d\n", x + 1, y + 1, min);

        totalCost += min;
        selected[y] = 1;
        edges++;
    }
    printf("\nMinimum Cost = %d\n", totalCost);

    return 0;
}
```

Output:


"C:\Users\Aditya\Downloads\
Enter number of vertices: 4
Enter the cost adjacency matrix:
Enter 0 if there is no edge.
0 10 6 8
10 0 12 14
6 12 0 18
8 14 18 0
Edges in Minimum Spanning Tree:
1 -- 3 = 6
1 -- 4 = 8
1 -- 2 = 10
Minimum Cost = 24